NATIONAL TEACHERS COLLEGE
School of Teacher Education
629 J. Nepomuceno St., Quiapo, Manila

Test Report

Choices	INTPUT	Expected Output	output	Result
1	([)]	Mismatch pair	Mismatch Pair	Pass
1	(2 +3	Error: Unclose Opening Bracket(s)	Error: Unclose Opening Bracket(s)	Pass
1	2 + 3)	Error: Unmatch closing Bracket ')'	Error: Unmatch closing Bracket ')'	Pass
1	5 / (3 - 3)	Division By zero	Division By zero	Pass
2	(12 + 3) * 2	12 3 + 2 * ; result 30	12 3 + 2 * ; result 30	Pass
2	8 + 2 * 5	8 2 5 * + ; result 18	8 2 5 * + ; result 18	Pass
2	20 / 5 / 2	20 5 / 2 / ; result 2	20 5 / 2 / ; result 2	Pass
2	10 - 3 - 2	10 3 - 2 - ; result 5	10 3 - 2 - ; result 5	Pass
2	{[2 + 3] * (4 + 1)}	2 3 + 4 1 + * ; result 25	2 3 + 4 1 + * ; result 25	Pass
2	7 / 2	7 2 / ; result 3.5	7 2 / ; result 3.5	Pass
2	0	0	Balance; 0	Pass
3	Run Stack Demonstration	pop() on empty stack returns	1. pop() on empty stack -> false; size = 0	Pass

		false; size = 0		
3	Run Stack Demonstration	push(10), push(20), peek() returns 20; size = 2	push(10), push(20), peek() -> 20; size = 2	Pass
3	Run Stack Demonstration	push(30) grows storage (cap 2 -> 4); size = 3; contents [10 20 30]	push(30) grows storage (cap 2 -> 4); size = 3; contents [10 20 30]	Pass
3	Run Stack Demonstration	Three pops return 30 20 10 (LIFO order)	Three pops return 30 20 10 (LIFO order)	Pass
3	Run Stack Demonstration	peek() on empty stack returns false, no crash, size = 0	peek() on empty stack returns false, no crash, size = 0	Pass

Reflection

The comprehensive test results confirm that the program functions seamlessly across all tested scenarios—flawlessly validating expression syntax, evaluating mathematical operations, and executing core dynamic stack operations while demonstrating robust error handling with a 100% pass rate.